Breaker Bay (Port / Industrial) Map Design

Executive Summary

Breaker Bay is a high-intensity **port-industrial battlefield** designed for **naval invasion, logistics warfare, and urban combat**. The terrain features massive cargo terminals, fortified seawalls, container yards, ship berths, industrial cranes, coastal roads, and dense port-adjacent housing blocks. The map emphasizes Obsidian Protocol's autonomous multi-domain tactics, sensor-fusion warfare, comms-relay networks, and logistics under heavy industrial congestion. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of port territory. Key features include naval landing zones, container-maze combat, elevated crane platforms, industrial choke corridors, and multi-route amphibious assault paths.

Design Goals & Gameplay

Naval Invasion & Coastal Assault

Breaker Bay supports coordinated naval landings, amphibious pushes, and ship-to-shore combat. Players must secure beachheads, docks, and industrial platforms.

Autonomy & Multi-Route Logistics

Autonomous AI chooses between pier landings, seawall breaches, container-yard infiltration, or elevated crane traversal. Units adapt to shifting cargo layouts and blocked roads.

Sensor & Information Warfare

Fog, sea spray, industrial smoke, and metal clutter distort sensors. Relay towers on cranes and rooftops restore clarity. Destroying relays creates fog-of-war pockets across port and sea.

Logistics & Resources

Cargo terminals, fuel depots, and ship berths serve as resupply nodes. Long supply lines across exposed docks and narrow industrial corridors force careful convoy protection.

Deployment & Progress

A fixed deployment budget (~30,000 points) forces players to choose between naval landing craft, heavy ground armor, or urban infantry. Blue deploys from the southwest landing pier; Red deploys from the northeast industrial command tower.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² port zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Landing Pier Command) — (0,0, 140×140 m Amphibious staging zone with naval craft and armored vehicles.

Red HQ (Industrial Command Tower) — (700, 700) 140×140 m Fortified port-authority tower with radar mast and AA emplacements.

Central Cargo Terminal — (500,450), 250×250 m Container stacks, forklifts, cranes, narrow industrial corridors.

Fuel & Logistics Depot — (250,800), 100×100 m Fuel tanks, pipelines, cargo trucks, supply crates.

Ship Berth Row — (800,300), 200×150 m Docked cargo ships, loading cranes, gangways.

Seawall & Breakwater — (0,450 → 400,900) Fortified wall with artillery positions and breach points.

Container Maze — (500,0 → 850,350) Dense stacked containers forming a close-range combat labyrinth.

Crane Ridge — (650,150), +260 m High-ground crane platforms with long-range sensor coverage.

Harbor Rooftops — (150,650), +240 m Comms relay tower and drone pads.

Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Seawall Breach Points

Critical for naval invasion; heavily fortified.

Container Maze

Close-range combat with tight corridors and ambush angles.

Cargo Terminal Lanes

Long straight industrial lanes ideal for armor and snipers.

Ship Berth Row

Mixed-range combat; naval units can support ground forces.

Sightlines

Crane Ridge provides long-range sensor coverage (~350 m). Fog and industrial smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW landing pier + naval staging area.

Red Deployment: NE industrial command tower + AA emplacements.

Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Concrete docks, seawalls, breakwaters, industrial yards, coastal roads.

Vegetation

Sparse palms, shrubs, industrial planters.

Buildings & Props

Military: Bunkers, AA guns, radar dishes, watchtowers. Industrial: Cranes, pipelines, generators, cargo containers. Logistics: Boats, trucks, forklifts, crates, fuel drums.

Effects

Fog, sea spray, smoke plumes, sparks, wind gusts.

Audio

Waves, machinery, horns, radio chatter, seagulls.

LOD & Budget

Asset Type	Count	Tri/Item	Total Tris
Seawall	3	200k	600k
Docks/Ca	4	80k	320k
Buildings	15	50k	750k
Roads/Pa	4	30k	120k
Vegetatio	150	2k	300k
Props	50	10k	500k
Total	—	—	**~2.59M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal AR template + Nreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out docks, crane terminal, seawall, AR anchor test.

2. Core Systems — Month 2 Roadblock, naval pathfinding, autonomous behaviors.

3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.

4. Content Fill — Month 4-5 Final models, textures, lighting, audio.

5. Gameplay & Polish — Month 6-7 Objectives, balance, VFX (fire, smoke, sea spray).

6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify naval landing logic, container-maze navigation, aerial flanking.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, multi-domain coordination, deployment budget enforcement, anchor drift, sensor accuracy.